

② Backtracking :- It is an algorithmic technique which uses brute force approach to find all possible solutions of any problem.

① Backtracking helps you in solving problems which are related to :-

1. Find all possible solutions.

2. Find all possible combinations

0 Backtracking :- Recursion +

(visit all the Nodes via which it is connected and then in the end once you are done visiting the Node q in that case you need to mark current Node visited as false).

eg:-

1. You are giving Rs 500 to your friend to do a certain task.
2. Your friend does that task for you. (Recursion f^{th} call),
3. Taking back Rs 500 from your friend (Backtracking)

1. $\text{vis}(\text{node}) = \text{true};$

2. Recursive fⁿ.

3. $\text{vis}(\text{node}) = \text{false}.$

① Trick for Backtracking:-

1. Do a certain operation depending on question...

2. Make Recursive 6^{th} call.

3. Undo operation which was performed in step No 1.

①

way 1

func (int index^{→0})

{

✓ 1. index++; // index = 1

✓ 2. func (index)
1

✓ 3. index--; // index = 0

}

① way 2

func (int index^{→0})

{

→ func (index + 1); ₁

$[0, 1, 2, 3]$ res. add(temp) 0, 1, 2, 3 (5)
 for (int i = index; i < nums.length; i++)
 {
 → temp.add(nums[i]);
 func(res, temp, nums, i+1);
 temp.remove(temp.size()-1);
 }

}

→ 0, 1, 2, 3, 4
 → 1, 2, 3, 4
 → 2, 3, 4
 → 3, 4